



React Native Master Guide — Level 4 (Complete Verbatim Edition)

Native Stack Navigation, Route Parameters, Persistent Bottom Tabs, Screen Focus (useFocusEffect) & Deep Linking

EXPO V54 • NATIVEWIND / TAILWIND CSS • ES6+



Level 4.1: Native Stack Navigation & Route Parameters

1. Mental Model: Web URLs vs. Mobile Stack Navigation

Web browsers lo manam `/home` nundi `/profile` ki velthe, patha page reload appoyi new page DOM loki vasthundi.

Kani Mobile lo navigation oka **Deck of Cards** 🃏 lanti **Stack**:

- **Push** (`navigate('Details')`): Patha screen paina kothha screen card slide ayyi vasthundi. Patha screen background lo active ga untundi (scroll position and state destroy avvavu!).
- **Pop** (`goBack()`): Top card remove ayyi kindunna patha screen direct ga kanipisthundi.

Stack Navigation (LIFO - Last In, First Out):

 Product Details	<--- Top of Stack (Current Screen)
 Home Feed Screen	<--- Underneath (State & scroll position preserved!)

2. Passing & Reading Route Parameters

```
// Screen A (Sender):
navigation.navigate('Details', {
  productId: '101',
  productName: 'iPhone 16 Pro',
  price: 134999,
});

// Screen B (Receiver):
const DetailsScreen = ({ route, navigation }) => {
  // Read params safely with optional chaining
  const { productId, productName, price } = route.params || {};

  return (
    <View>
      <Text>{productName} - ₹{price}</Text>
      <Pressable onPress={() => navigation.goBack()}>
        <Text>Go Back 🏠</Text>
      </Pressable>
    </View>
  );
};
```

Code Example: Interactive Multi-Screen Stack System

```
import React, { useState } from 'react';
import { View, Text, Pressable, FlatList } from 'react-native';
import { SafeAreaView, SafeAreaView } from 'react-native-safe-area-context';
import { StatusBar } from 'expo-status-bar';

// Sample Catalog Data
const PRODUCTS = [
  { id: '1', name: 'MacBook Pro M4', category: 'Laptops', price: 199999, desc: '16-core GPU, Liquid Retina XDR display, 36GB u',
  { id: '2', name: 'Sony WH-1000XM5', category: 'Audio', price: 29999, desc: 'Industry-leading noise cancellation, 30hr batter',
  { id: '3', name: 'Apple Watch Ultra 2', category: 'Wearables', price: 89999, desc: 'Rugged titanium case, precision dual-fre'
];

// =====
// 📱 SCREEN 1: Home Feed Screen (Screen A)
// =====
const HomeScreen = ({ onNavigate }) => {
  return (
    <View className="flex-1 px-5 pt-3">
      { /* Title */ }
      <View className="mb-4">
        <Text className="text-2xl font-black text-sky-400">Native Stack 🕒</Text>
        <Text className="text-slate-400 text-xs">Tap an item to push Details Screen to the stack</Text>
      </View>

      { /* Product List */ }
      <FlatList
        data={PRODUCTS}
        keyExtractor={({item}) => item.id}
        ItemSeparatorComponent={() => <View className="h-3" />}
        renderItem={({ item }) => (
          <Pressable
            android_ripple={{ color: '#38bdf820' }}
            onPress={() => onNavigate('Details', item)} // Passing Route Params!
            className="bg-slate-900 border border-slate-800 p-4 rounded-2xl flex-row justify-between items-center active:borde
          >
            <View className="flex-1 pr-3">
              <Text className="text-white font-bold text-base">{item.name}</Text>
              <Text className="text-slate-400 text-xs mt-0.5">{item.category}</Text>
              <Text className="text-emerald-400 font-extrabold text-sm mt-1.5">
                ₹{item.price.toLocaleString('en-IN')}
              </Text>
            </View>

            <View className="bg-sky-500/10 px-3 py-1.5 rounded-xl border border-sky-500/20">
              <Text className="text-sky-400 font-bold text-xs">View →</Text>
            </View>
          </Pressable>
        )}
      />
    </View>
  );
};

// =====
// 📱 SCREEN 2: Details Screen (Screen B)
// =====
const DetailsScreen = ({ routeParams, onGoBack }) => {
  const { name, category, price, desc, id } = routeParams || {};

  return (
    <View className="flex-1 px-5 pt-3 justify-between pb-6">
      <View>
        { /* Back Button Header */ }
        <Pressable
          hitSlop={15}
          onPress={onGoBack}
          className="flex-row items-center gap-2 mb-6 self-start bg-slate-900 px-3.5 py-2 rounded-xl border border-slate-800 a
        >
          <Text className="text-sky-400 font-extrabold text-sm">⬅️ Back to Feed</Text>
        </Pressable>
      </View>
    </View>
  );
};
```

```

    {/* Product Card */}
    <View className="bg-slate-900 border border-slate-800 p-6 rounded-3xl gap-3 shadow-2xl">
      <View className="bg-sky-500/20 self-start px-3 py-1 rounded-full border border-sky-500/30">
        <Text className="text-sky-400 text-xs font-bold uppercase">{category}</Text>
      </View>

      <Text className="text-2xl font-black text-white">{name}</Text>

      <Text className="text-3xl font-extrabold text-emerald-400">
        ₹{price?.toLocaleString('en-IN')}
      </Text>

      <View className="h-[1px] bg-slate-800 my-2" />

      <Text className="text-slate-300 text-sm leading-relaxed">{desc}</Text>

      <View className="bg-slate-950 p-3 rounded-xl border border-slate-800 mt-2">
        <Text className="text-slate-500 text-[11px] font-mono">
          📦 Received Route Param ID: #{id}
        </Text>
      </View>
    </View>

    {/* Checkout Action Button */}
    <Pressable
      android_ripple={{ color: 'ffffff' }}
      onPress={() => alert(`Purchasing ${name} for ₹${price?.toLocaleString('en-IN')}!`)}
      className="w-full bg-emerald-500 active:bg-emerald-600 p-4 rounded-xl items-center"
    >
      <Text className="text-slate-950 font-black text-base">Buy Now 🛒</Text>
    </Pressable>
  </View>
);
};

// =====
// 🚀 ROOT APP (Stack Navigator Engine)
// =====
const App = () => {
  const [currentScreen, setCurrentScreen] = useState('Home');
  const [routeParams, setRouteParams] = useState(null);

  const navigate = (screenName, params = null) => {
    setRouteParams(params);
    setCurrentScreen(screenName);
  };

  const goBack = () => {
    setRouteParams(null);
    setCurrentScreen('Home');
  };

  return (
    <SafeAreaView>
      <StatusBar style="light" backgroundColor="#090d16" />
      <SafeAreaView className="flex-1 bg-slate-950">
        {currentScreen === 'Home' ? (
          <HomeScreen onNavigate={navigate} />
        ) : (
          <DetailsScreen routeParams={routeParams} onGoBack={goBack} />
        )}
      </SafeAreaView>
    </SafeAreaView>
  );
};

export default App;

```

Telugu + English Explanation:

- **Stack Mechanism:** Home screen nundi item tap cheyagane, `navigate('Details', item)` call ayyi details screen open avthundi. Back button nokkithe `goBack()` call ayyi patha screen loki instantly return avtham.
- **Route Params Flow:** Screen A nundi `item` object ni pass chestham, Screen B lo `routeParams` access chesi data ni render chestham.
- **State Preservation:** Mobile lo stack navigation valla user details screen ki velli back vachina, home screen lo unna scroll position and data reset avvavu.

Quick Check:

Q: Which method removes the top card from the stack?

A: B) `navigation.goBack()`. It pops the top screen card and reveals the previous screen beneath it!

🔗 Level 4.2 & 4.3: Bottom Tabs, Screen Focus (useFocusEffect) & Deep Linking

1. Bottom Tab Navigation

Mobile apps lo persistent app-wide navigation kosam **Bottom Tabs** (like Instagram, Swiggy, YouTube) vaadatham.

- Prathi tab ki **Icon**, **Active Tint Color**, and **Notification Badges** (`tabBarBadge: 4`) untayi.
- Tabs switch chesthunappudu screens background memory lo cached ga untayi (data loss avvadu).

2. 🚨 The Big Mobile Gotcha: Why useEffect FAILS on Screen Return!

Web lo vere page nundi back vasthe `useEffect` malli run avthundi. **Kani mobile Navigation Stack / Tabs lo Screen unmount avvadu (background lo alage untundi).**

The Solution: useFocusEffect & useIsFocused: React Navigation provide chese `useFocusEffect` hook screen prathi sari user kanta padinappudu (focus loki vachinappudu) execute avthundi!

User leaves Screen A → Screen A sleeps in background (Still mounted)
User returns to Screen A → `useEffect()` does NOT run ❌
→ `useFocusEffect()` RUNS IMMEDIATELY! ✅ (Refreshes Cart / Wallet)

3. Deep Linking (`myapp://...`)

WhatsApp or SMS lo link click chesinappudu (e.g., `myapp://product/101` or `https://myapp.com/orders`), direct ga app open ayyi aa specific screen loki velladaniki **Deep Linking** vaadatharu (Configured in `app.json` via `"scheme": "myapp"`).

Code Example: Interactive Bottom Tabs & Live Focus Sensor

```
import React, { useState, useEffect } from 'react';
import { View, Text, Pressable, FlatList } from 'react-native';
import { SafeAreaView, SafeAreaView } from 'react-native-safe-area-context';
import { StatusBar } from 'expo-status-bar';

// =====
// 🏠 TAB 1: Home Feed Screen
// =====
const HomeTab = ({ visitCount }) => {
  return (
    <View className="flex-1 px-5 pt-4">
      <Text className="text-2xl font-black text-sky-400 mb-1">Explore Store 🛒</Text>
      <Text className="text-slate-400 text-xs mb-6">Bottom Tab Navigation & Focus Lifecycle</Text>

      {/* Focus Lifecycle Tracker Card */}
      <View className="bg-slate-900 border border-slate-800 p-5 rounded-3xl mb-4 gap-2 shadow-xl">
        <View className="flex-row items-center gap-2">
          <View className="w-2.5 h-2.5 rounded-full bg-emerald-400 animate-pulse" />
          <Text className="text-white font-bold text-sm">useFocusEffect Active</Text>
        </View>
        <Text className="text-slate-400 text-xs leading-relaxed">
          Switch to "Cart" tab and switch back to "Home". Notice how the focus counter automatically increments!
        </Text>
        <View className="bg-slate-950 p-3 rounded-xl border border-slate-800 mt-2 flex-row justify-between items-center">
          <Text className="text-slate-400 text-xs">Screen Focused:</Text>
          <Text className="text-emerald-400 font-extrabold text-sm">{visitCount} times</Text>
        </View>
      </View>
    </View>
  );
};

// =====
// 🛒 TAB 2: Cart Screen (with Badge Counter)
// =====
const CartTab = ({ cartItems, onClearCart }) => {
  return (
    <View className="flex-1 px-5 pt-4 justify-between pb-4">
      <View>
        <Text className="text-2xl font-black text-sky-400 mb-1">Shopping Cart 🛒</Text>
        <Text className="text-slate-400 text-xs mb-6">Persistent Tab State</Text>

        <View className="bg-slate-900 border border-slate-800 p-5 rounded-3xl gap-3">
          <Text className="text-white font-bold text-base">Items in Bag ({cartItems.length})</Text>
          {cartItems.map((item, idx) => (
            <View key={idx} className="bg-slate-950 p-3 rounded-xl border border-slate-800 flex-row justify-between">
              <Text className="text-slate-200 text-xs font-semibold">{item.name}</Text>
              <Text className="text-emerald-400 text-xs font-bold">₹{item.price}</Text>
            </View>
          ))}
        </View>
      </View>

      <Pressable
        onPress={onClearCart}
        className="bg-slate-800 active:bg-red-500/20 border border-slate-700 p-3.5 rounded-xl items-center">
        <Text className="text-red-400 font-bold text-xs">Clear Cart</Text>
      </Pressable>
    </View>
  );
};

// =====
// 🧑 TAB 3: Profile Screen
// =====
const ProfileTab = () => {
  return (
    <View className="flex-1 px-5 pt-4">
      <Text className="text-2xl font-black text-sky-400 mb-1">Account & Settings ⚙️</Text>
      <Text className="text-slate-400 text-xs mb-6">Deep Linking: myapp://profile</Text>
    </View>
  );
};
```

```

<View className="bg-slate-900 border border-slate-800 p-6 rounded-3xl items-center gap-3">
  <View className="w-20 h-20 rounded-full bg-sky-500/20 border border-sky-500/30 items-center justify-center">
    <Text className="text-4xl">👤</Text>
  </View>
  <Text className="text-white font-extrabold text-lg">Kiran Kumar</Text>
  <Text className="text-slate-400 text-xs">React Native Mobile Engineer</Text>

  <View className="w-full bg-slate-950 p-3.5 rounded-xl border border-slate-800 mt-2">
    <Text className="text-sky-400 font-bold text-xs mb-0.5">Universal Deep Link Scheme:</Text>
    <Text className="text-slate-500 text-[11px] font-mono">myapp://user/profile?ref=share</Text>
  </View>
</View>
</View>
);
};

// =====
// 🚀 ROOT APP WITH PERSISTENT BOTTOM TAB BAR
// =====
const App = () => {
  const [activeTab, setActiveTab] = useState('Home');
  const [homeFocusCount, setHomeFocusCount] = useState(1);
  const [cartItems, setCartItems] = useState([
    { name: 'MacBook Pro M4', price: '1,99,999' },
    { name: 'AirPods Pro', price: '24,900' },
  ]);

  // Simulating useFocusEffect when activeTab changes to Home!
  const handleTabChange = (tabName) => {
    if (tabName === 'Home' && activeTab !== 'Home') {
      setHomeFocusCount((prev) => prev + 1); // Triggers on screen focus!
    }
    setActiveTab(tabName);
  };

  const tabs = [
    { id: 'Home', label: 'Feed', icon: '🏠' },
    { id: 'Cart', label: 'Cart', icon: '🛒', badge: cartItems.length },
    { id: 'Profile', label: 'Profile', icon: '👤' },
  ];

  return (
    <SafeAreaView>
      <StatusBar style="light" backgroundColor="#090d16" />
      <SafeAreaView className="flex-1 bg-slate-950 justify-between">

        {/* Active Screen Body */}
        <View className="flex-1">
          {activeTab === 'Home' && <HomeTab visitCount={homeFocusCount} />}
          {activeTab === 'Cart' && <CartTab cartItems={cartItems} onClearCart={() => setCartItems([])} />}
          {activeTab === 'Profile' && <ProfileTab />}
        </View>

        {/* 📱 PERSISTENT BOTTOM NAVIGATION TAB BAR */}
        <View className="bg-slate-900 border-t border-slate-800/80 px-6 py-3 flex-row justify-around items-center">
          {tabs.map((tab) => {
            const isActive = activeTab === tab.id;
            return (
              <Pressable
                key={tab.id}
                onPress={() => handleTabChange(tab.id)}
                className="items-center relative py-1 px-4">
                >
                {/* Notification Badge */}
                {tab.badge > 0 && (
                  <View className="absolute top-0 right-3 bg-red-500 w-4 h-4 rounded-full items-center justify-center z-10">
                    <Text className="text-white text-[9px] font-black">{tab.badge}</Text>
                  </View>
                )}

                <Text className="text-xl mb-0.5">{tab.icon}</Text>
                <Text

```

```

        className={`text-[11px] font-bold ${
          isActive ? 'text-sky-400' : 'text-slate-500'
        }}
      >
        {tab.label}
      </Text>
      {isActive && (
        <View className="w-1.5 h-1.5 rounded-full bg-sky-400 mt-1" />
      )}
    </Pressable>
  );
  })}
</View>

</SafeAreaView>
</SafeAreaProvider>

);
};

export default App;

```

👤 Telugu + English Explanation:

- **useFocusEffect vs useEffect:** User vere tab ki velli malli Home tab ki vachinappudu `useEffect` run avvadu. Kani `useFocusEffect` prathi sari screen focus loki ragane execute avthundi (Wallet balance update cheyadaniki or fresh notifications load cheyadaniki idi use chestham).
- **Tab Badges:** Cart icon paina unde red badge (`tabBarBadge: 2`) user items add/remove chesinappudu live ga update avthundi.
- **Deep Linking:** `myapp://profile` lanti URL scheme unte, outside apps (WhatsApp / Chrome) nundi link click cheyagane mana mobile app direct ga open avthundi.